

Strukturplan für die Bachelorarbeit

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August 14, 2026

General Information

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Studiengang: Informatik
Sprache: Englisch
Titel: Implementation and Visualization of a Post-Quantum Cryptosystem
Using the Example of the McEliece Cryptosystem

Research Question

How can the McEliece cryptosystem be implemented and visualized within CrypTool Online to improve the understanding of code-based post-quantum cryptography for students?

Executive Summary

The emergence of quantum computing poses a significant threat to currently deployed public-key cryptographic systems such as RSA and Elliptic Curve Cryptography. Algorithms like Shor's algorithm are expected to compromise the security assumptions of these systems once sufficiently powerful quantum computers become available. Consequently, post-quantum cryptography has become an important research and development area within modern information security.

Among the various approaches to post-quantum cryptography, the McEliece cryptosystem represents one of the oldest and most extensively studied code-based cryptographic systems. Despite decades of cryptanalytic research, no practical attacks against properly parameterized variants of the system are known. However, due to its strong mathematical foundations in coding theory, the McEliece cryptosystem is often perceived as difficult to understand, especially for students encountering post-quantum cryptography for the first time.

This thesis addresses the challenge of making the McEliece cryptosystem more accessible for educational purposes. Rather than focusing on advanced mathematical proofs or cryptanalytic research, the thesis investigates how the essential concepts and processes of the cryptosystem can be presented in a comprehensible and interactive manner.

The primary objective is the design and implementation of an educational visualization tool based on CrypTool Online (CTO). The work includes the identification of relevant theoretical concepts, the development of a didactic visualization approach, the design of a software architecture suitable for educational use, and the implementation of a CrypTool Online plugin that visualizes the key generation, encryption, and decryption processes of the McEliece cryptosystem.

The resulting artifact aims to support students in understanding the relationship between coding theory and post-quantum cryptography while demonstrating how interactive visualizations can improve the learning process in complex security-related domains.

The thesis contributes both a practical software implementation and a didactic framework for teaching code-based post-quantum cryptography.

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1 Introduction

Ziel: max. 2 Seiten.

1.1 Motivation

- Bedrohung klassischer Kryptographie durch Quantencomputer
- Relevanz von PQC
- Schwierigkeit des Verständnisses codebasierter Verfahren
- Bedarf an Lehr- und Lernwerkzeug

1.2 Research Question

Vorstellung der Forschungsfrage

1.3 Structure of the Thesis

Kurze Beschreibung aller Kapitel

2 Fundamentals of Cryptography

*Knapp halten – Zielgruppe (Absolvent*innen des eigenen Studiengangs) kennt die Grundlagen bereits. Fokus nur auf das, was für Kapitel 4 (McEliece) und die Abgrenzung zu klassischen Verfahren wirklich gebraucht wird.*

2.1 Symmetric and Asymmetric Cryptography

- Grundprinzipien
- Key Exchange
- Public-Key-Verfahren

2.2 Security Concepts

- Confidentiality
- Integrity
- Authenticity

2.3 Computational Hardness

- One-Way Functions
- Trapdoor Functions

2.4 Public-Key Cryptography

- RSA
- ECC
- Typische Sicherheitsannahmen

3 Quantum Computing and Post-Quantum Cryptography

3.1 Quantum Computing Fundamentals and Impact on Classical Cryptography

- Qubits, Superposition, Quantum Advantage
- Shor's Algorithm – Auswirkungen auf RSA und ECC
- Grover's Algorithm – Auswirkungen auf symmetrische Verfahren

3.2 Need for Post-Quantum Cryptography

- Store Now, Decrypt Later
- Migration Challenges

3.3 Overview of PQC Families

- Lattice-based
- Code-based
- Hash-based
- Multivariate
- Isogeny-based

3.4 Position of McEliece within PQC

- Historische Entwicklung
- NIST-Prozess
- Classic McEliece

4 The McEliece Cryptosystem

4.1 Error-Correcting Codes

- Motivation
- Kommunikationsfehler
- Fehlerkorrektur

4.2 Linear Codes

- Codewords
- Generator Matrix
- Parity Check Matrix

4.3 Goppa Codes

- Grundidee
- Eigenschaften
- Fehlerkorrekturfähigkeit

4.4 Structure of the McEliece Cryptosystem

- Private Key
- Public Key
- Message
- Error Vector

4.5 Key Generation

- Generator Matrix
- Scrambling Matrix
- Permutation Matrix
- Public Key Construction

4.6 Encryption

- Encoding
- Error Addition

4.7 Decryption

- Decoding
- Error Correction

4.8 Security Assumption

- General Decoding Problem
- Schwierigkeit zufälliger linearer Codes

4.9 Educational Abstraction

5 CrypTool Online Architecture

5.1 Introduction to CrypTool

- CrypTool Projekt
- CrypTool Online

5.2 Architecture of CrypTool Online

- Workflow-Konzept
- Plugins
- Datenflüsse

5.3 Existing Educational Components

- Analyse bestehender CTO-Plugins

5.4 Related Work

- Vergleich mit bestehenden Lehrmitteln zu code-basierter/Post-Quanten-Kryptographie
- Abgrenzung: was macht diese Arbeit anders/zusätzlich?

6 Educational and Didactic Concept

6.1 Learning Objectives

- PQC verstehen
- McEliece einordnen können
- Schlüsselabläufe nachvollziehen können

6.2 Target Audience Analysis

- Bachelor-Studierende
- Grundkenntnisse Kryptographie

6.3 Educational Challenges

- Abstrakte Mathematik
- Hohe Komplexität

6.4 Visualization Principles

- Cognitive Load Theory
- Progressive Disclosure
- Step-by-Step Visualization

6.5 Concept of the Learning Tool

- Lernszenarien
- Benutzerfluss
- Interaktionen

7 Requirements

7.1 Problem Statement

- McEliece ist mathematisch anspruchsvoll
- Vorhandene Literatur richtet sich häufig an Experten
- Fehlende interaktive Lernwerkzeuge

7.2 Objectives

- Entwicklung eines CTO-Plugins
- Visualisierung zentraler Abläufe
- Unterstützung des Lernprozesses

7.3 Scope

- Abgrenzung der Arbeit
- Festlegung der unterstützten McEliece-Parametergrößen (Lehr-/Demo-Parameter vs. reale/klassische Sicherheitsparameter)

7.4 Functional Requirements

7.5 Non-Functional Requirements

7.6 Requirements for the McEliece Plugin

- Fachliche Anforderungen
- Didaktische Anforderungen
- Technische Anforderungen

8 Design of the McEliece Learning Plugin

8.1 System Design

- Architekturdiagramm

8.2 User Interface Design

- Layout
- Navigation
- Interaktion

8.3 Visualization Design

- Prozessdiagramme
- Matrixdarstellung
- Animationen

8.4 Data Flow Design

- Eingaben
- Ausgaben
- Zwischenzustände

9 Implementation

9.1 Development Environment

- CTO Framework
- Programmiersprache
- Tools

9.2 Implementation of Key Generation

- Technische Umsetzung

9.3 Implementation of Encryption

- Technische Umsetzung

9.4 Implementation of Decryption

- Technische Umsetzung

9.5 Visualization Components

- Animationen
- Grafiken
- Interaktive Elemente

9.6 Challenges and Solutions

- Technische Probleme
- Didaktische Probleme

10 Evaluation

10.1 Evaluation Method

- Experten-Review und/oder kleine Nutzerstudie mit Kommiliton*innen
- Abgleich mit den Lernzielen aus Abschnitt 6.1

10.2 Study Design / Participants

10.3 Results

10.4 Discussion of Results

10.5 Limitations of the Evaluation

11 Conclusion and Future Work

11.1 Summary of Contributions

11.2 Answering the Research Question

11.3 Limitations

11.4 Future Work

12 Sources

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- Bundesamt für Sicherheit in der Informationstechnik (BSI): Quantentechnologien und Post-Quanten-Kryptografie
https://www.bsi.bund.de/EN/Themen/Unternehmen-und-Organisationen/Informationen-und-Empfehlungen/Quantentechnologien-und-Post-Quanten-Kryptografie/quantentechnologien-und-post-quanten-kryptografie_node.html
- BSI: Kryptografie quantensicher gestalten
https://www.bsi.bund.de/SharedDocs/Downloads/DE/BSI/Publikationen/Broschueren/Kryptografie-quantensicher-gestalten.pdf?__blob=publicationFile&v=6

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